Assignment Topic :

Programming_Fundamental_Lab _02

Submitted by :

Muhammad Zain Ul Abedein

Department :

BSCS__1st.

Roll No :

33

Section :

Section__D

Submitted To :

SIR ASAD.

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Task: 1

Type and save the Following Program on Visual studio. Run the Program and observe their out Put.

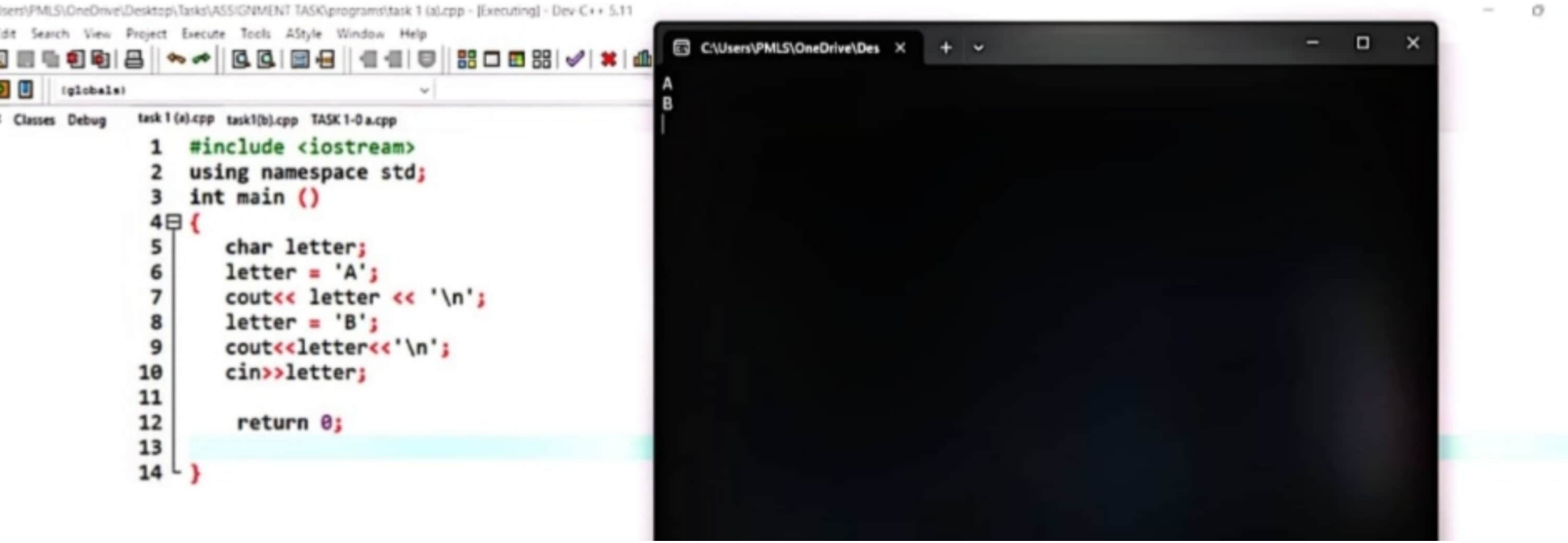
(i)

```
#include <iostream>
using namespace std;
int main ( ) {
char letter;
letter = 'A';
cout <<letter << '\n';
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}
```

Out Put :

A

B


Search View Project Execute Tools AStyle Window Help
(globals)
Classes Debug
task 1 (a).cpp task1(b).cpp TASK 1-0 a.cpp
1 #include <iostream>
2 using namespace std;
3 int main ()
4 {
5 char letter;
6 letter = 'A';
7 cout<< letter << '\n';
8 letter = 'B';
9 cout<<letter<<'\n';
10 cin>>letter;
11
12 return 0;
13
14 }
C:\Users\PMLS\OneDrive\Des
A
B

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11)

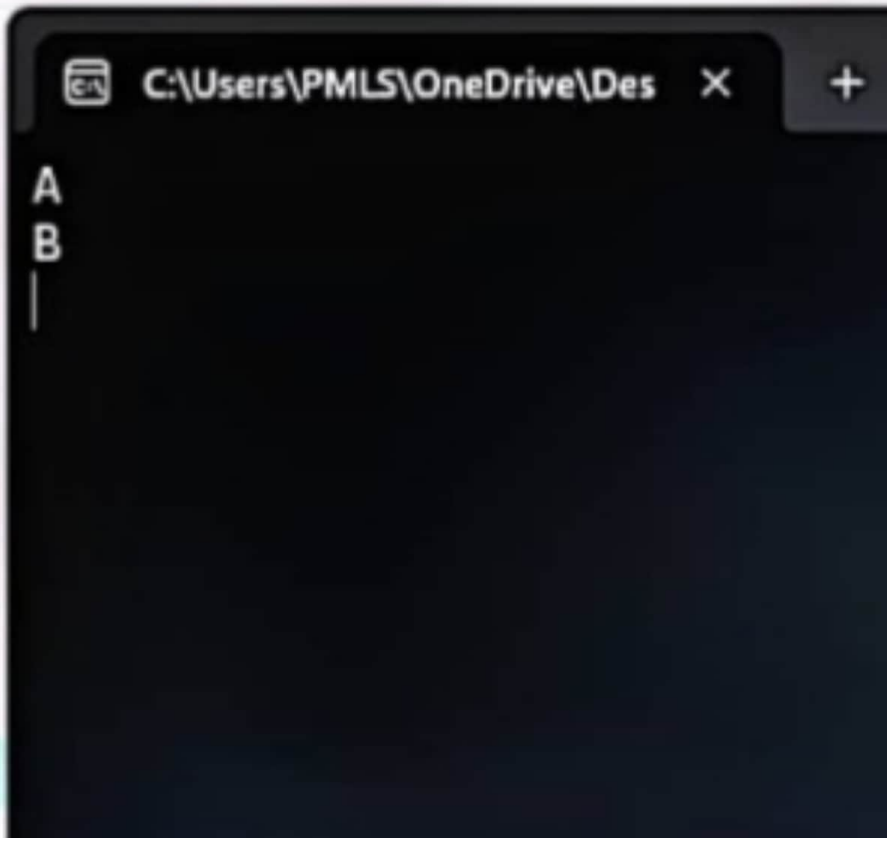
```
#include <iostream>
using namespace std;
int main ( ) {
char letter;
letter = 65;
cout << letter << '\n';
letter = 66;
cout << letter << '\n';
cin >> letter;
return 0;
}
```

Out Put :

A
B

[*] task 1 (a).cpp task1(b).cpp

```cpp
#include <iostream>
using namespace std;
int main () {
char letter;
letter = 65;
cout <<letter<<'\n';
letter = 66;
cout <<letter<<'\n';
cin>>letter;
    return 0;
}
```



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iii)

```
#include <iostream>
using namespace std;
int main ( ) {
    int letter;
    letter = 'A';
cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
cin >> letter;
    return 0; }
```

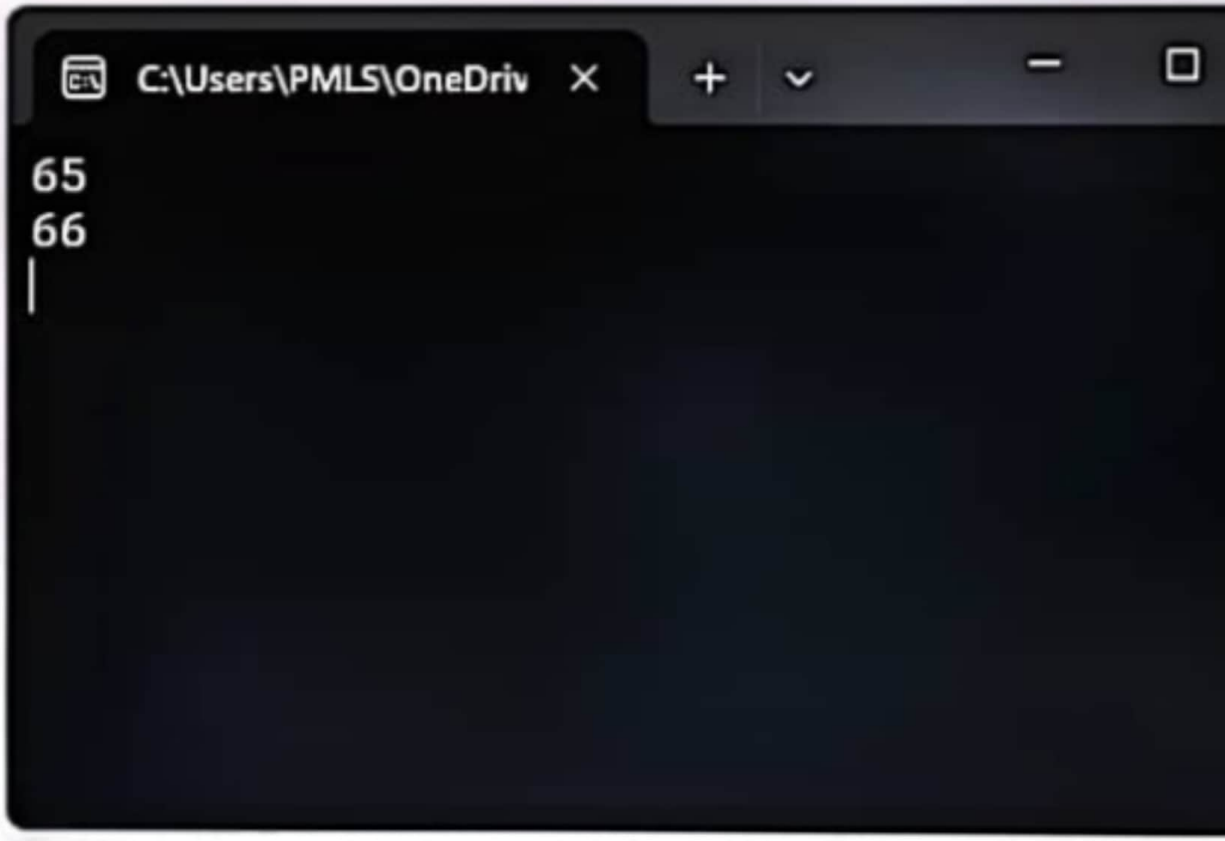
output : 65
66

task 1 (a).cpp task1(b).cpp TASK 1-0 a.cpp

```cpp
#include <iostream>
using namespace std;
int main ()
{
   int letter;
   letter = 'A';
   cout<< letter<<'\n';
   letter = 'B';
   cout<<letter<<'\n';
   cin>>letter;

     return 0;

}
```



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Task: 2

Take two integers input From the user apply arithmatic operation on (+, -, *, \) Print on screen.

```
#include <iostream>
using namespace std;
int main ( ) {
int a, b ;
  a = 4 ;
  b = 2 ;
cout << a+b<< '\n';
cout<< a-b<< '\n';
cout<<a*<< '\n';
cout<< a /b ;
  return 0 ;
}
```

Out Put :

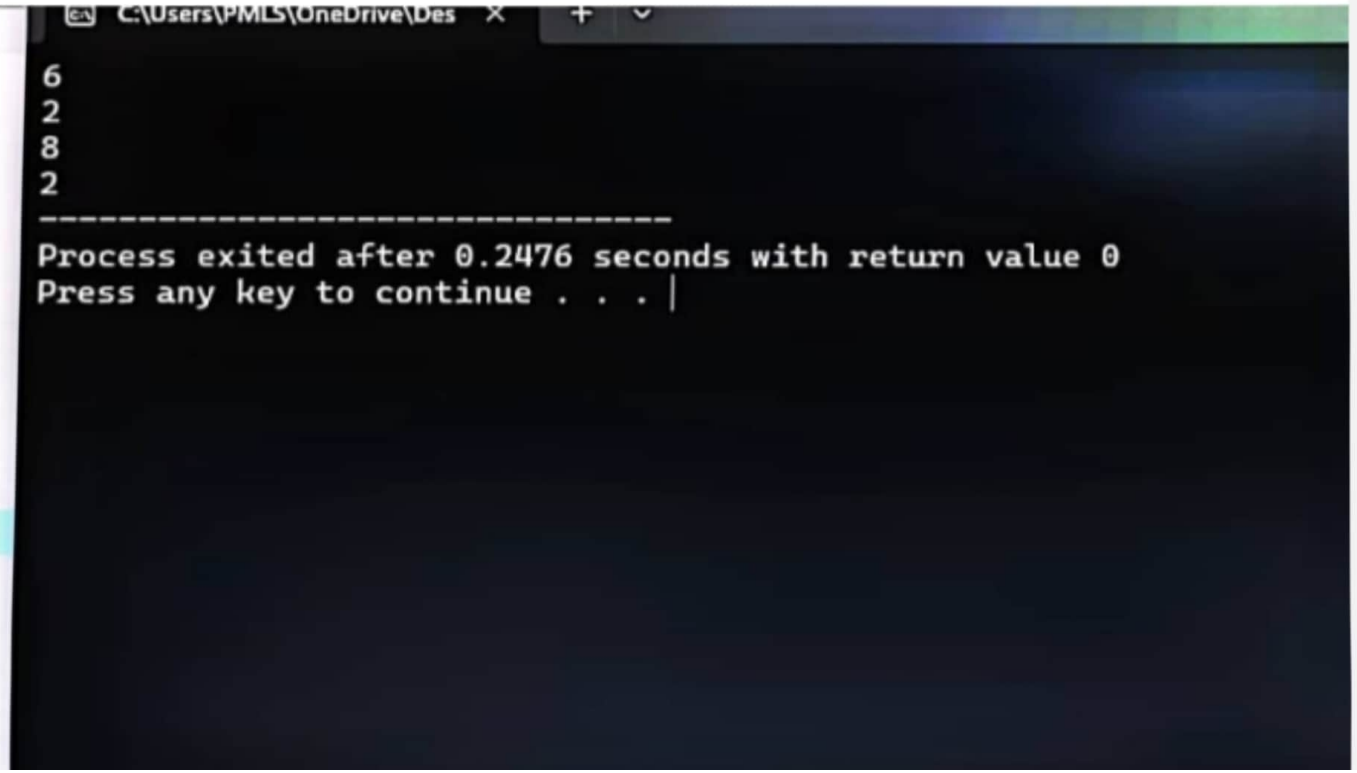
```
6
2
8
2
```

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task 1 (a).cpp task1(b).cpp TASK 1-0 a.cpp

```cpp
#include <iostream>
using namespace std;
int main ()
{
 int a,b;
  a = 4;
  b = 2;
 cout<<a+b<<'\n';
 cout<<a-b<<'\n';
 cout<<a*b<<'\n';
 cout<<a/b;
 return 0;

}
```



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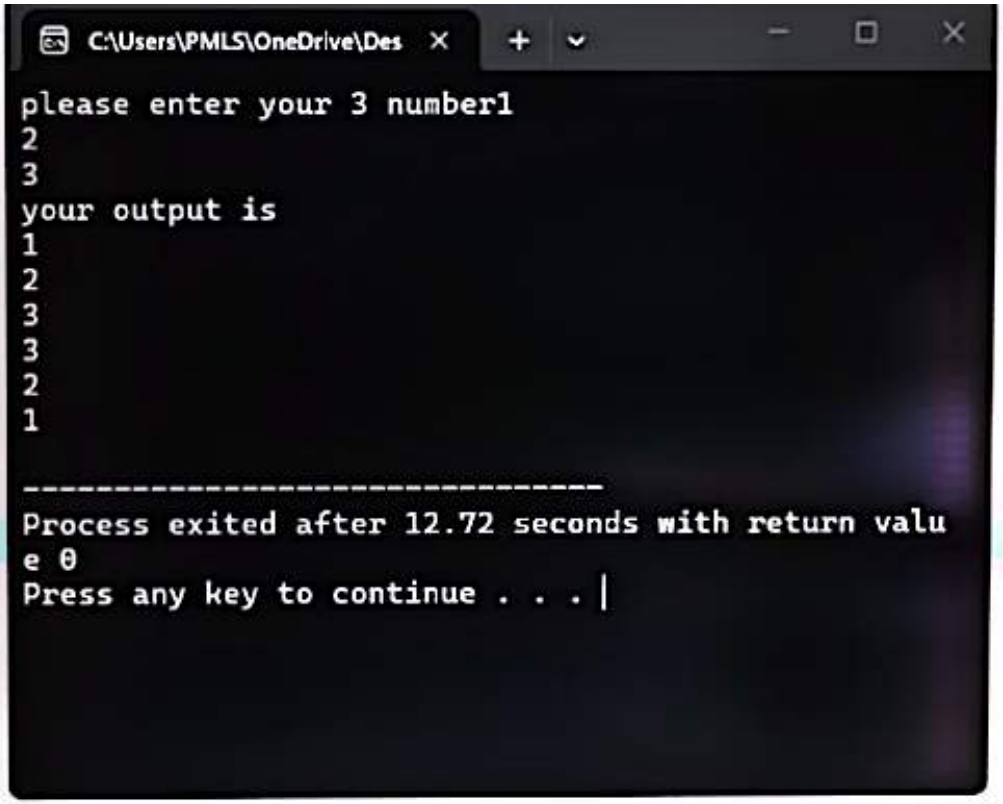
Task # 03.

```
#include <iostream>
using namespace std;
    int main () {
int val= 1;
int val= 2;
int val 3;
cout << " Please enter your 3 numbers:";
cin >> val1 >> val2 >> val3;
cout << "Your output is:"endl;
 cout<<val1<<"\n"<<val2<<"\n<<val3<<"\n";
 cout<<val3<<"\n"<<val2<<"\n"<< val3<<"\n";
    return0;
    }
```

output on next page

Untitled1.cpp

```cpp
#include <iostream>
using namespace std;
int main ()
{
int val1;
int val2;
int val3;
cout << "please enter your 3 number";
cin>> val1>>val2>>val3;
cout << "your output is"<<endl;
cout <<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
cout <<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
    return 0;
}
```



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Task # 04.

```
#include <iostream>
using namespace std;
int main ()
{
    double x1, x2, y1, y2, d;
    cin>> x1>>y1>>x2>>y2;
    d= sqrt((x2-x1)*(x2-x1)+(y2-y1)*(y2-y1));
    cout << "Distance = " << d;
    return 0;
}
```

If input is
$x_1 = 1, x_2 = 3$
$y_1 = 2, y_2 = 4$
So output is

Distance = 2.828

```cpp
#include <iostream>
#include <cmath>
using namespace std;

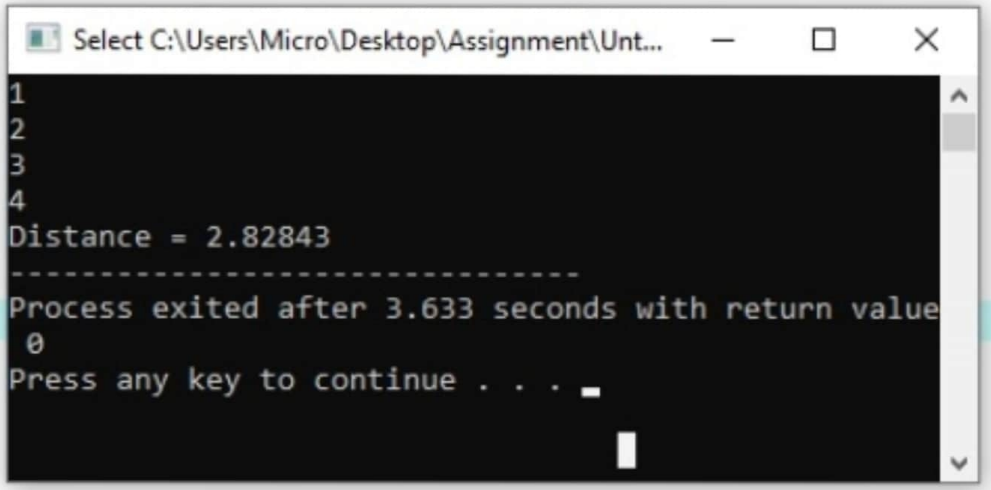
int main()
{
    double x1, x2, y1, y2, d;

    cin >> x1 >> y1 >> x2 >> y2;

    d = sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));

    cout << "Distance = " << d;

    return 0;
}
```



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Task # 5' - Part-a.

Area = $\pi \cdot r^2$.

```
#include <iostream>
using namespace std;
    int main ()
{
    float. area, radius = 5';
    area = 3.1416 * radius * radius;
    cout << "area of a circle:" << area;
    return 0;
    }
```

area of circle = 78.54

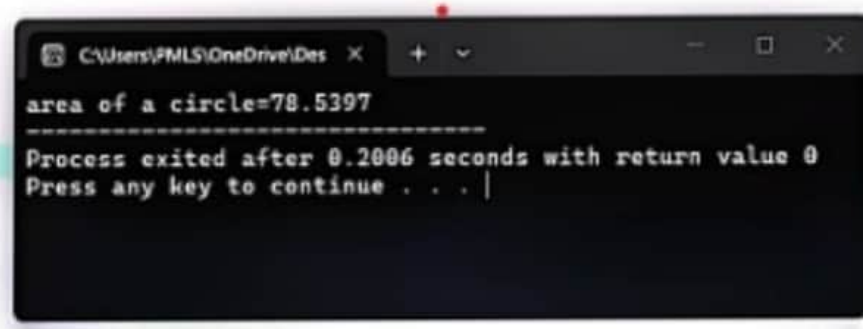
Part-#-b.

Area of rectangle = length * width.

```
#include <iostream>
using namespace std;
    int main () {
int area, length = 8, width = 10;
area = length * width;
cout << "Area of rectangle" << area;
    return 0;
    }
```

Area of rectangl : 80

```cpp
#include <iostream>
using namespace std;
int main ()
{
float area,radius=5;
area = 3.14159*radius*radius;
cout <<"area of a circle="<<area;
    return 0;
}
```



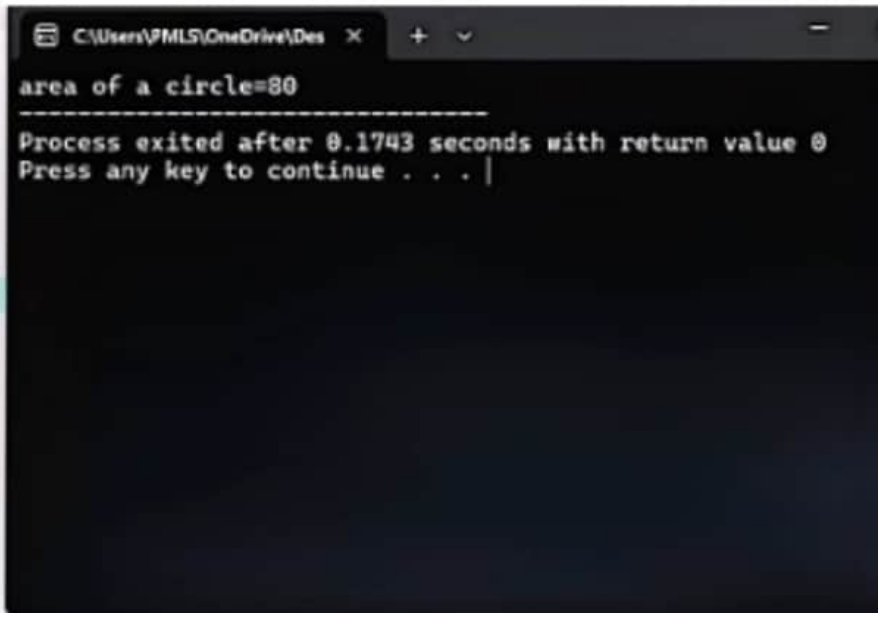
Compile Log | Debug | Find Results | Close

```
mpilation results...
-------
Errors: 0
Warnings: 0
Output Filename: C:\Users\PMLS\OneDrive\Desktop\Tasks\ASSIGNMENT TASK\programs\Untitled1.exe
Output Size: 1.832423210144044 MiB
Compilation Time: 0.55s
```

Scanned with CamScanner

Untitled1.cpp

```cpp
#include <iostream>
using namespace std;
int main ()
{
int area,length =8,width =10;
area = length*width;
cout << "area of a circle="<<area;
return 0;
}
```



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Part – c.

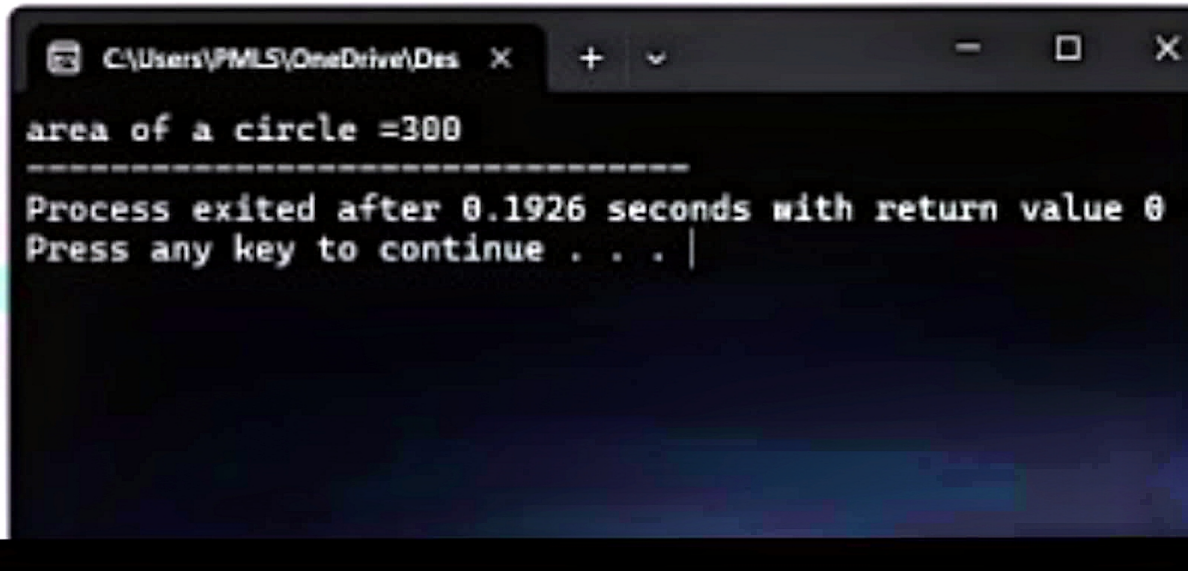
Area of Triangle.

:- (area = base * height * .5.)

```
#include <iostream>
using namespace std;
   int main ()
{
 int area, base=20, height=30;
area = base * height * .5;
cout << "Area of Triangle:" << area;
   return 0;
   }
```

Area of Triangle : 300

```cpp
#include <iostream>
using namespace std;
int main ()
{
int area,base = 20,height = 30;
area = base*height*.5;
cout << "area of a circle ="<<area;
return 0;
}
```



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Task # 06.

```
#include <iostream>
using namespace std;
int main ()
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double items = 3.95;
double subtotal = item1 + item2 + item3 + item4 + items;
    double tax = subtotal * 0.06;
    double total = subtotal + tax;

cout << "Item 1: $" << item1 << endl;
cout << "Item 2: $" << item2 << endl;
cout << "Item 3: $" << item3 << endl;
cout << "Item 4: $" << item4 << endl;
cout << "Items: $" << items << endl;
    cout << "Subtotal: $" << subtotal << endl;
    cout << "Sales Tax (6%): $" << tax << endl;
    cout << "Total Bill" << total << endl;
    return 0;
}
```

Item 1 = $ 12.95
Item 2 = $ 24.95
Item 3 = $6.95
Item 4 = $14.95
Item 5 = $3.95

Subtotal = $63.75
Sales Tax = $3.025
Total Bill = $ 67.575.

```
3   int main ()
4   {
5       double item1 =12.95, item2 =24.95, item3 =6.95, item4 =14.95, item5 =3.95;
6       double subtotal = item1 + item2 + item3 + item4 + item5;
7       double tax =subtotal * 0.06;
8       double total = subtotal + tax;
9       cout << "price of item 1 = $"<< item1 << endl;
10      cout << " price of item 2= $" << item2 << endl;
11      cout << "price of item 3= $"<< item3 << endl;
12      cout << "price of item4= $"<< item4<< endl;
13      cout << "price of item5= $"<<item5<< endl;
14      cout << "subtotal= $" << subtotal << endl;
15      cout << "sales tax =$"<< tax << endl;
16      cout << "total =$"<< total << endl;
17      return 0;
```

C:\Users\PAK COPMUTERS\Desktop\task6.exe

```
price of item 1 = $12.95
 price of item 2= $24.95
price of item 3= $6.95
price of item4= $14.95
price of item5= $3.95
subtotal= $63.75
sales tax =$3.825
total =$67.575
```

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Bonus Task:

```
#include <iostream>
using namespace std;
    int main ( ) {
    int num ;
cout <<"Enter a five-digit number:";
  cin >> num;
  cout << num / 10000 <<"    ";
  cout<<( num / 1000 ) %. 10 <<"    ";
  cout<<( num / 100 ) %. 10 <<"    ";
  cout<<( num /10) %. 10 <<"    ";
  cout << num %. 10 << endl;
    return 0; }
```

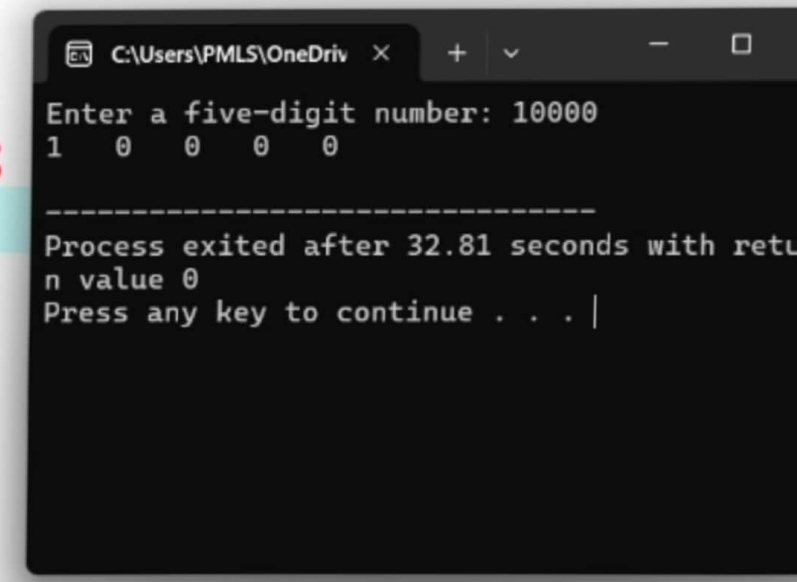
OūPut:

```
Enter a five-digit numbr: 10000
1 0 0 0 0
```

home task 1.cpp | [*] Bonus task lab 2.cpp

```cpp
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a five-digit number: ";
    cin >> num;
    cout << num / 10000 << "    ";
    cout << (num / 1000) % 10 << "    ";
    cout << (num / 100) % 10 << "    ";
    cout << (num / 10) % 10 << "    ";
    cout << num % 10 << endl;

    return 0;
}
```



Compile Log | Debug | Find Results | Close

```
mpilation results...
------
Errors: 0
```

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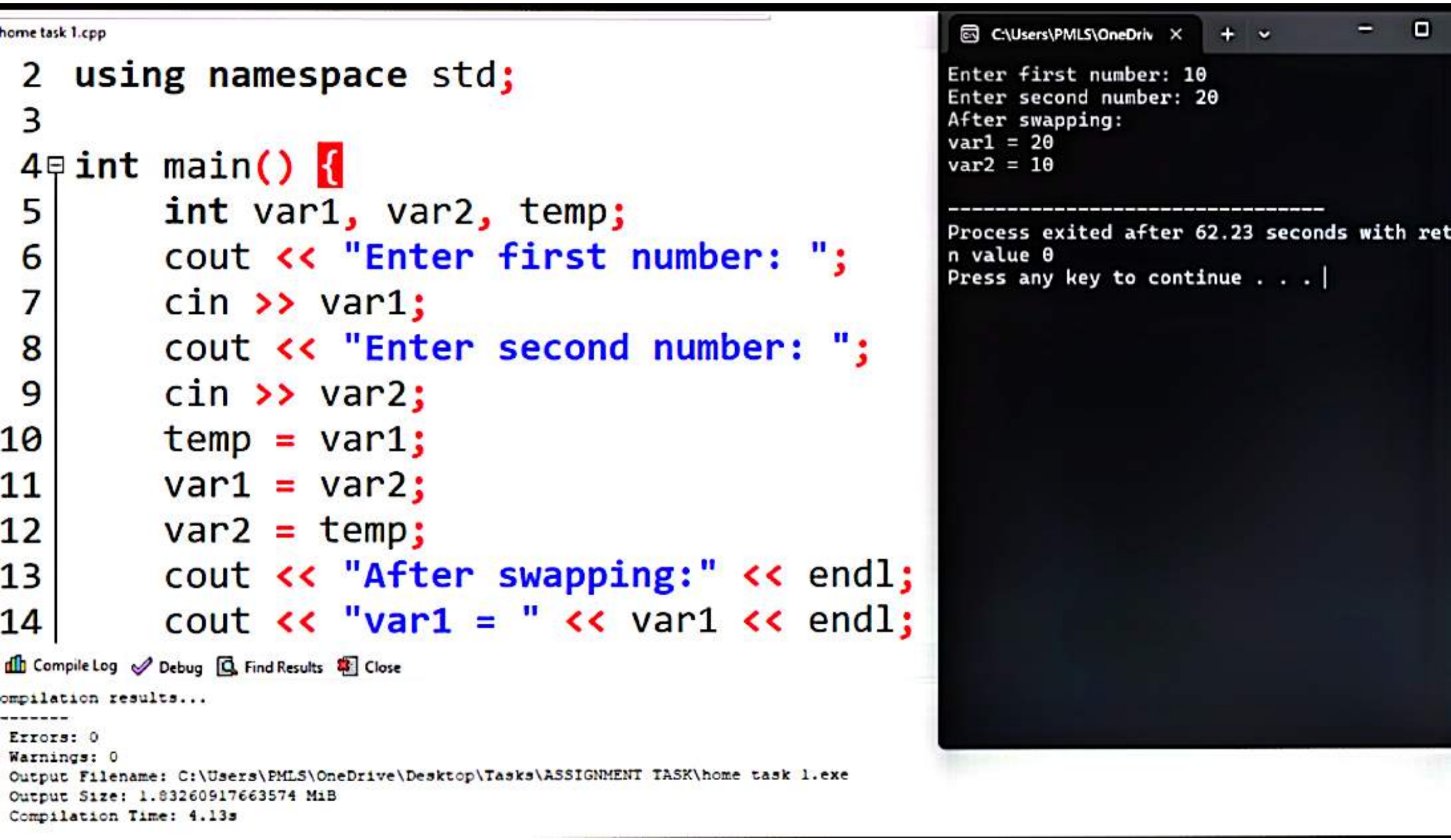
Home Task: 1

```
#include <iostream>
using name space std;
int main ( ) {
int var1, var2, temp;
cout << "Enter first number:";
 cin >> var1;
cout << "Enter second number:";
 cin >> var2;
 temp = var1;
 var1 = var2;
 var2 = temp;
 cout << "After swapping:" << endl;
cout << "Var1 =" << var1 << endl;
```

Output

```
Enter First number: 10
Enter second number: 20
After swapping:
 var1 = 20
 var2 = 10
```

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home task 1.cpp
using namespace std;
int main() {
int var1, var2, temp;
cout << "Enter first number: ";
cin >> var1;
cout << "Enter second number: ";
cin >> var2;
temp = var1;
var1 = var2;
var2 = temp;
cout << "After swapping:" << endl;
cout << "var1 = " << var1 << endl;
Compile Log
Debug
Find Results
Close
ompilation results...
Errors: 0
Warnings: 0
Output Filename: C:\Users\PMLS\OneDrive\Desktop\Tasks\ASSIGNMENT TASK\home task 1.exe
Output Size: 1.83260917663574 MiB
Compilation Time: 4.13s
C:\Users\PMLS\OneDriv
Enter first number: 10
Enter second number: 20
After swapping:
var1 = 20
var2 = 10
Process exited after 62.23 seconds with ret
n value 0
Press any key to continue . . .

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HomeTask : 02

```
#include <iostream>
using namespace std;
int main() {
    char ch;
    cout << "Enter a small alphabet: ";
    cin >> ch;
    ch = ch - 32;
    cout << "uppercase alphabet: " << ch << endl;

    return 0;
}
```

Output:

```
Enter a small alphabet : a
uppercase alphabet : A
```

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```
3
4  int main() {
5      char ch;
6
7      cout << "Enter a small alphabet: ";
8      cin >> ch;
9
10     ch = ch - 32;
11
12     cout << "Uppercase alphabet: " << ch << endl;
13
14     return 0;
15 }
```

```
C:\Users\PMLS\OneDriv
Enter a small alphabet: a
Uppercase alphabet: A

--------------------------------
Process exited after 24.39 seconds with retur
n value 0
Press any key to continue . . .
```

Compile Log | Debug | Find Results | Close

```
Compilation results...
--------
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\PMLS\OneDrive\Desktop\Tasks\ASSIGNMENT TASK\home task 1.exe
- Output Size: 1.83260917663574 MiB
- Compilation Time: 0.70s
```

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PF LAB 01 COMPLETED ✔️

THANK YOU,😊

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